CNENGR115 Computers and Control

Revision Questions

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Example question 1

By making a number of assumptions concerning friction etc., the movement of a motor vehicle can be represented by the following standard 1st order Transfer Function,

$$X = \frac{G}{Ts + 1} U$$

where X is the vehicle speed (mph) and U is the engine force. In this model, the engine force is represented as a scaled value between 0 and 1.0, where 1.0 implies maximum force, 0.5 half of this and so on.

Example question 1

Using the Transfer Function model above, and $T = 5.2$ seconds and $G = 87$, sketch the response of vehicle speed to a step in engine force from zero to 0.5 units. Label your sketch to define the *steady state speed* and *time constant* (stating all the relevant numerical values and units). Suggest two limitations of this model.

Example question 1

Draw a block diagram of an *integral controller* applied to the 1st order Transfer Function stated above. Add the following labels: vehicle speed, control input, control model and set point. Use the rules of block diagram analysis to find the closed-loop Transfer Function. Find the steady state gain of the closed-loop Transfer Function and comment on the significance of your answer.

The control algorithm is $U = \frac{K_I}{s}(D - X)$ where K_I is the gain and D the set point.

Example question 2

Consider the following differential equation, where x is the output, $u(t)$ the input and a_2, b_1, b_2 are parameters.

$$\frac{d^2x}{dt^2} + a_2 x = b_1 \frac{du(t)}{dt} + b_2 u(t)$$

Develop the Transfer Function of the above differential equation.

Example question 2

An engineer proposes to design the following control algorithm, where D is the set point, and K_P, K_V are control gains:

$$U = K_P(D - X) + K_V X$$

Draw a block diagram of the control system stated above, as applied to the model found in the first part of this question.

Example question 3

- Know your control objectives and how they relate to a particular control situation.
  - Stability
  - Tracking
  - Transient behaviour
  - Disturbances